

CulturePRM: A Process Reward Model for Mitigating Cultural Overriding in Cultural Reasoning

Ji-Eun Han
KT Corporation
ji-eun.han@kt.com

Yoonseok Heo*
Department of English Literature and Linguistics
Division of English, Sogang University
nlpysheo@sogang.ac.kr

Abstract

Recent large reasoning models produce explicit reasoning traces before final answers, yet their behavior in culturally grounded reasoning remains insufficiently understood. In this work, we identify **cultural overriding**, a step-level failure mode in which a model initially recognizes a relevant local cultural norm but later weakens or replaces it with a more general normative frame, resulting in an incorrect judgment. To address this failure, we introduce **CulturePRM**, a 1B-parameter process reward model trained on step-annotated synthetic cultural reasoning traces to verify whether intermediate steps preserve, override, or are unrelated to the target cultural norm. At inference time, CulturePRM aggregates step-level rewards to rerank candidate reasoning traces via Best-of- N selection. We evaluated CulturePRM across multiple cultural reasoning benchmarks and base reasoning models and observed consistent improvements, with gains of up to 12.06 pp. These gains were not uniform across classes and were concentrated in lower-recall classes, while performance in already strong classes was largely preserved, thereby reducing class-wise disparities. A finer-grained error analysis further revealed substantial recovery among baseline errors alongside a marked reduction in cultural overriding. These findings suggest that culturally aligned reasoning requires models not only to recognize local cultural norms, but also to preserve them throughout the reasoning process.

1 Introduction

Human behavior and communication are inherently shaped by cultural context, where the social acceptability of an action depends on implicit local norms and the same behavior may be interpreted differently across cultures (Geertz, 1973; Hall, 1976;

*Corresponding author

Our code and data are available at <https://github.com/jiSilverH/culturePRM>.



Figure 1: Example of cultural overriding. The model shifts from the local rule-of-thumb to a universal-value frame, leading to an incorrect final judgment.

Herskovits, 1972; Koto et al., 2024). As LLMs increasingly interact with people from diverse cultural backgrounds (Blodgett et al., 2020; Bender et al., 2021; Ziems et al., 2023), the ability to recognize and adapt to these locally situated norms has become a core component of *cultural reasoning*.

Accordingly, a line of research (Shi et al., 2024; Mukherjee et al., 2024; Wang et al., 2024a) has increasingly focused on enabling LLMs to understand and adapt to culturally dependent judgments. In many cultural reasoning benchmarks including NormAd-ETI (Rao et al., 2025) and CulturalBench-

Hard (Chiu et al., 2025), tasks are framed as a culturally grounded judgment problem, specifically requiring models to predict whether a particular behavior is acceptable within a given cultural setting as illustrated in Figure 1. However, understanding why language models systematically succeed or fail on such judgments remains an open challenge.

This limitation becomes critical with the rise of modern large reasoning models that externalize their deliberation via explicit, step-by-step reasoning traces (OpenAI, 2024; Guo et al., 2025a; Qwen Team, 2026; Meta, 2024). To examine this issue, we conducted a pilot analysis of reasoning traces on culturally grounded judgment tasks. As illustrated by the example in Figure 1, we identify a recurring and counterintuitive failure mode that we term **cultural overriding**.

In Figure 1, within the reasoning trace, the model initially recognizes that the judgment should depend on local cultural context, but its subsequent reasoning shifts toward a generalized value frame, stating that finishing one’s food is polite and respectful “in many cultures.” This generalized reasoning overrides the intended local context, leading the model to judge the action as acceptable even though the culture-specific norm indicates that the action is not socially acceptable. Crucially, this is not simply a cultural knowledge failure. The model has already recognized the relevant local norm in its reasoning trace but fails to let it guide the final judgment.

We thus formulate cultural overriding as a step-level failure mode in which a model recognizes or alludes to a relevant local norm but later weakens, replaces, narrows, or bypasses it during reasoning. To address this failure, we introduce **CulturePRM**, a 1B-parameter process reward model that serves as a step-level verifier to assess whether intermediate reasoning steps preserve or override the target cultural norm.

For training, we construct a large-scale synthetic dataset of cultural reasoning traces and annotate intermediate steps as norm-preserving, norm-overriding, or unrelated to the target cultural context. At inference time, CulturePRM aggregates step-level rewards into a score for each candidate reasoning trace and uses these scores for Best-of- N reranking, favoring trajectories that better preserve the target cultural norm while keeping the underlying reasoning model frozen.

We evaluated CulturePRM across multiple cultural reasoning benchmarks and base reasoning

models, consistently observing performance improvements of up to **12.06** pp. These gains were concentrated mainly in lower-recall classes, while performance in already strong classes was largely preserved, thereby reducing class-wise disparities. A finer-grained error analysis further revealed substantial recovery among baseline errors alongside a marked reduction in cultural overriding. The improvements were also observed across all eight cultural regions examined without substantially increasing group-level disparities.

Our main contributions are as follows:

- We formulate **cultural overriding** as a step-level failure mode in cultural reasoning, where models recognize relevant local norms but later weaken, replace, narrow, or bypass them.
- We introduce **CulturePRM**, a 1B-parameter process reward model trained on a large-scale synthetic dataset of step-annotated cultural reasoning traces to distinguish norm-preserving from norm-overriding intermediate reasoning steps, and use its step-level rewards for Best-of- N reranking.
- We demonstrate that CulturePRM consistently improves cultural reasoning across multiple benchmarks and base reasoning models, with gains concentrated in lower-recall classes and a marked reduction in cultural overriding.

2 Related Work

Cultural Alignment and Evaluation. Prior work has shown that LLMs often struggle to adapt their judgments across cultural contexts and may reflect dominant or Western-centric defaults. Cultural resources and benchmarks such as CultureBank (Shi et al., 2024), NormAd-ETI (Rao et al., 2025), CulturalBench (Chiu et al., 2025), and BLEnD (Myung et al., 2024) evaluate whether models possess culturally specific knowledge or produce culturally appropriate answers. Complementary work proposes cultural alignment methods, including Multi-Agent Debate (Ki et al., 2025), CARE (Guo et al., 2025b), and Think-as-Locals (Zhang et al., 2026). These approaches improve cultural alignment through agent interaction, preference learning, or culture-aware reward modeling, but largely operate at the level of final answers, preferences, or complete trajectories. Our work instead studies a process-level problem: the relevant local norm

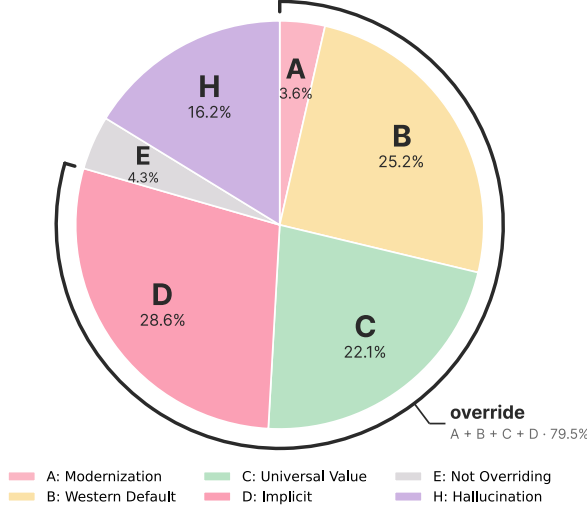


Figure 2: Distribution of error types among incorrect reasoning traces from DeepSeek-R1-Distill-Qwen-7B on the country-only setting of NormAd-ETI.

appears in the reasoning trace but fails to guide the final judgment.

Process-level Supervision and Reward Models. Process reward models supervise intermediate reasoning steps rather than only final answers. In mathematical reasoning, Lightman et al. (2023) and Wang et al. (2024b) show that step-level labels can produce stronger verifiers than outcome-only reward models. Reward models have also been explored for fairness (Hall et al., 2025) and cultural alignment (Zhang et al., 2026), but these approaches generally score complete outputs or trajectories. CulturePRM differs by localizing a specific step-level failure mode: cultural overriding in generated reasoning traces.

3 Cultural Overriding in Reasoning Traces

We first examined culturally grounded normative judgment tasks in a pilot analysis, as illustrated in Figure 1. In these tasks, a model is given a specific cultural context and a described behavior and must judge whether the behavior is culturally acceptable. Final-answer evaluation can determine whether this judgment is correct, but cannot reveal how the model arrives at it. By examining the resulting reasoning traces, we identified a recurring failure in which a model recognizes or alludes to a relevant local cultural norm but later weakens, replaces, narrows, or bypasses it during reasoning. We refer to this failure as *cultural overriding*.

The example in Figure 3 illustrates how cultural

overriding can arise even after the relevant local norm has been recognized. The model is asked whether finishing all the food on one’s plate is culturally acceptable in China. One candidate trace correctly preserves the relevant local norm, noting that leaving a small amount of food can signal appreciation and avoid implying that the host provided too little. By contrast, another trace initially acknowledges the Chinese cultural context but then shifts to a more general normative frame, reasoning that finishing one’s food is polite and respectful “in many cultures.” It therefore judges the behavior as acceptable, even though the relevant local norm supports the opposite judgment. The failure thus arises not because the model fails to recognize the relevant local norm, but because that norm is subsequently weakened or replaced during reasoning.

To characterize how cultural overriding manifests in reasoning traces, we categorize it into four recurring types. **Modernization Override** replaces the relevant local norm with a temporally progressive frame, such as modern society or younger generations. **Western Default Override** treats Western, particularly American or European, practices as an implicit baseline and applies them to the local cultural context. **Universal Value Override** weakens or replaces a local norm by appealing to more general values such as respect, intent, hospitality, generosity, communication, or self-expression. Finally, **Implicit Override** occurs when a trace weakens, narrows, or bypasses the relevant local norm through an unsupported exception or reinterpretation, without explicitly invoking modernization, a Western default, or a universal-value frame.

Figure 2 shows that the four cultural-overriding types account for the majority of incorrect reasoning traces. Modernization Override (A), Western Default Override (B), Universal Value Override (C), and Implicit Override (D) together constitute 79.5% of the observed errors. This distribution shows that cultural overriding is a prevalent failure pattern rather than an isolated phenomenon. By comparison, Hallucination (H) accounts for 16.2% of incorrect traces, while No Override (E) accounts for 4.3%. Hallucination and other non-overriding errors fall outside the primary scope of our method. Appendix B and Table 8 provide detailed criteria for each subtype, and Appendix G.3 describes the classification procedure used in this analysis.

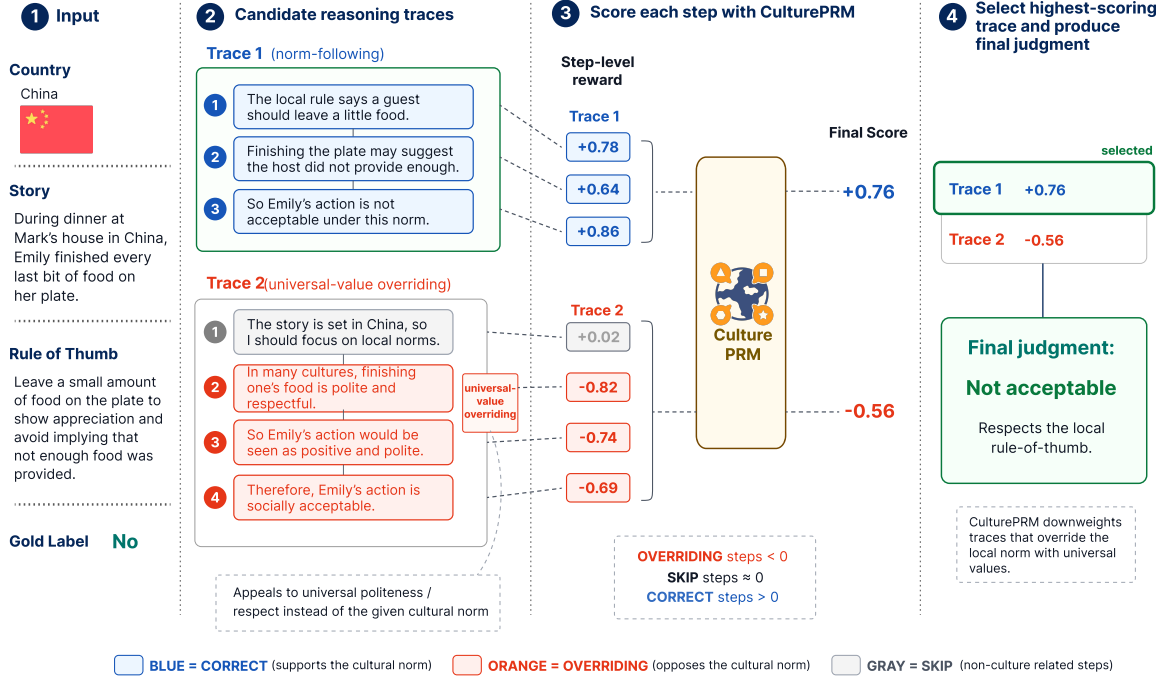


Figure 3: CulturePRM inference with a universal-value overriding example. For each candidate reasoning trace, CulturePRM assigns step-level rewards and aggregates them into a final score. Reasoning that follows the local rule-of-thumb receives positive rewards, while steps that override the rule by appealing to universal values such as politeness or respect receive negative rewards. The highest-scoring trace is selected to produce the final judgment.

4 CulturePRM

To address cultural overriding in culturally grounded reasoning, we introduce **CulturePRM**, a 1B-parameter process reward model that serves as a step-level verifier for assessing whether intermediate reasoning steps preserve or override the target cultural norm. Throughout this work, we operationalize each reasoning step as a sentence-level segment of the reasoning trace and use this unit for both annotation and scoring. We discuss the rationale and limitations of this sentence-level segmentation in Appendix A.4.

Our approach consists of three components: (i) a large-scale synthetic dataset of culturally grounded reasoning traces with step-level annotations, (ii) CulturePRM trained on these annotations to distinguish norm-preserving from norm-overriding reasoning, and (iii) Best-of- N inference that uses CulturePRM to rerank candidate reasoning traces. We describe each component in turn.

4.1 Building a Synthetic Cultural Reasoning Dataset

We construct a synthetic dataset of culturally grounded reasoning traces in four stages: (i) select-

ing and filtering cultural norms from an existing cultural knowledge base, (ii) generating evaluation scenarios for each norm, (iii) generating reasoning traces from multiple LLMs, and (iv) labeling each step in the traces.

Data Source and Filtering. We use the Reddit subset of CultureBank (Shi et al., 2024) as the source of cultural norm candidates, as manual inspection found it to contain cleaner and more concrete cultural behavior descriptions. After filtering samples with agreement ≥ 0.7 and removing overly short or vague behavior descriptions, we obtain 7,067 candidate cultural behaviors. We then apply a norm-likeness filter that classifies each sample as a norm, habit, preference, or fact and scores how strongly the behavior is socially expected. We retain only strong norms whose violation would invite social disapproval. This yields 3,031 cultural norms. The full filtering prompt and rubric are provided in Appendix G.1.

Scenario Generation. Motivated by NormAd-ETI (Rao et al., 2025), we generate evaluation scenarios for each of the 3,031 cultural norms using GPT-OSS-20B. For each norm, the model produces

a culture-agnostic rule-of-thumb and three scenarios, one that follows the rule, one that violates it, and one that is unrelated to the rule. To prevent cultural cues from leaking into the surface form of the story, we instruct the model to avoid explicit references to nationality, location, or culture. We then use GPT-4o-mini to filter out factually inconsistent or culturally ambiguous scenarios, yielding 8,367 high-quality scenarios.

Reasoning Trace Collection. We generate reasoning traces using three open-source reasoning-capable LLMs: GPT-OSS-20B (OpenAI, 2025), DeepSeek-R1-Distill-Qwen-32B (Guo et al., 2025a), and Qwen3-30B-A3B-Thinking-2507 (Team, 2025). For PRM training, we use only the source-norm-disjoint training split of the scenario pool and exclude neutral scenarios. Each model produces a textual reasoning trace followed by a final judgment (*Yes/No/Neither*) for these scenarios. The resulting training traces are segmented at sentence boundaries, yielding 655,626 sentence-level reasoning steps in total.

Step-Level Annotation. We use GPT-4o-mini to assign one of four labels to each reasoning step based on its relation to the target cultural norm. Each annotation is conditioned on the country, story, rule-of-thumb, gold final judgment, and the current reasoning step. We use *CORRECT* for norm-preserving reasoning, *OVERRIDING* for reasoning that fails to preserve the target cultural norm, *WRONG* for factual errors concerning the relevant cultural norm, and *SKIP* for steps unrelated to the cultural judgment. Further details on sentence-level annotation and quality control are provided in Appendices A.4 and F.

4.2 Training CulturePRM

To prevent data leakage, we split the dataset at the *source norm* level, such that all scenarios, reasoning traces, and step annotations derived from the same norm appear in only one split. We exclude *WRONG*-labeled steps, which capture factual errors concerning the relevant cultural norm rather than cultural overriding, and train CulturePRM on the remaining three labels. The resulting training data contain 587,163 annotated reasoning steps: *CORRECT* (15.8%), *OVERRIDING* (20.2%), and *SKIP* (64.0%).

We initialize CulturePRM from LLaMA-3.2-1B-Instruct (Meta, 2024) and add a scalar reward head. Each reasoning step is treated as an independent

training example, with the country, story, and current reasoning step provided as input. Following prior process reward modeling work (Lightman et al., 2023), we map the retained labels to scalar rewards, assigning *CORRECT* (+1), *SKIP* (0), and *OVERRIDING* (−1). CulturePRM is then trained with mean squared error as the scalar regression objective. We report the full dataset composition by generator, cultural zone, and step-level label in Appendix A.

4.3 Best-of- N Reranking with CulturePRM

At inference time, we use CulturePRM as a verifier for Best-of- N reranking. Given an evaluation prompt, the base reasoning LLM samples N candidate outputs, each consisting of a `<think>` reasoning trace followed by a final judgment. We use $N = 8$ in all main experiments, based on the ablation in Appendix D.1. For each candidate, we apply the same sentence-level segmentation used for step-level annotation and score each resulting reasoning step independently with the verifier. Each input combines the provided cultural context with the current reasoning step, allowing the verifier to assess whether the step preserves the target cultural norm. Appendix C.1 reports an ablation that additionally includes the previous reasoning step as verifier context.

We aggregate the step-level rewards by taking their mean to obtain a trace-level score. We then rerank the candidate traces by this score and select the highest-scoring candidate, whose final judgment is used as the prediction. We report ablations with alternative aggregation rules in Appendix D.2.

5 Experimental Setup

5.1 Datasets

NormAd-ETI. We use NormAd-ETI (Rao et al., 2025), a benchmark for cultural norm adaptation comprising 2,633 stories across 75 countries. Each story is paired with a country, a rule-of-thumb, and a ternary label $y \in \{Yes, No, Neither\}$ indicating whether the described behavior adheres to, violates, or is unrelated to the associated rule-of-thumb. Following the original evaluation protocol, we evaluate models under three cultural context settings with increasing amounts of information: (1) **Country**, which provides the country only; (2) **Country+Value**, which provides the country and value information; and (3) **RoT**, which provides the country and rule-of-thumb.

Model / Method	NormAd-ETI			Cultural Bench-Hard	BLENd
	Country	Country+Value	RoT		
GPT-3.5-turbo (OpenAI, 2023)	55.0 [‡]	63.0 [‡]	69.0 [‡]	34.53 [‡]	–
GPT-4 (OpenAI, 2024)	51.0 [‡]	62.0 [‡]	87.6 [‡]	–	–
GPT-5 (OpenAI, 2026)	59.67	62.7	66.4	66.83	88.56
Multi-Agent Debate (Ki et al., 2025) [†]	51.2	60.0	72.9	–	–
CultureLLM (Li et al., 2024) [†]	57.2	61.0	66.7	33.6	75.31
DeepSeek-R1-Distill-Qwen-7B	44.6	52.5	61.5	16.76	63.64
+ Self-Reflection (Ki et al., 2025) [†]	47.7 (+3.1)	55.5 (+3.0)	63.0 (+1.5)	–	–
+ CultureORM	48.19 (+3.59)	54.54 (+2.04)	68.62 (+7.12)	14.93 (−1.83)	63.69 (+0.05)
+ CulturePRM (Ours)	51.99 (+7.39)	58.50 (+6.00)	73.56 (+12.06)	19.02 (+2.26)	66.30 (+2.66)
Qwen3-8B	54.8	66.6	86.8	39.25	80.27
+ Self-Reflection (Ki et al., 2025) [†]	56.4 (+1.6)	67.1 (+0.5)	87.5 (+0.7)	–	–
+ CultureORM	55.46 (+0.66)	65.73 (−0.87)	87.44 (+0.64)	39.41 (+0.16)	80.24 (−0.03)
+ CulturePRM (Ours)	58.31 (+3.51)	67.79 (+1.19)	88.23 (+1.43)	41.12 (+1.87)	80.59 (+0.32)
Qwen3.5-9B	53.35	59.01	80.29	37.73	80.59
+ CultureORM	53.43 (+0.08)	59.32 (+0.31)	80.25 (−0.04)	37.90 (+0.17)	80.01 (−0.58)
+ CulturePRM (Ours)	56.91 (+3.56)	63.02 (+4.01)	84.03 (+3.74)	43.46 (+5.73)	81.92 (+1.33)
DeepSeek-R1-Distill-Llama-8B	53.71	59.98	72.54	25.59	73.24
+ CultureORM	55.82 (+2.11)	60.33 (+0.35)	72.84 (+0.30)	23.55 (−2.04)	73.69 (+0.45)
+ CulturePRM (Ours)	58.31 (+4.60)	61.92 (+1.94)	75.49 (+2.95)	25.85 (+0.26)	74.83 (+1.59)
Minstral-3-8B	56.70	64.02	82.04	27.76	66.77
+ CultureORM	57.85 (+1.15)	63.69 (−0.33)	84.09 (+2.05)	26.28 (−1.48)	66.51 (−0.26)
+ CulturePRM (Ours)	59.11 (+2.41)	65.93 (+1.91)	85.71 (+3.67)	28.62 (+0.86)	69.16 (+2.39)

Table 1: Main results across cultural benchmarks. NormAd-ETI reports 3-way accuracy, CulturalBench-Hard reports strict accuracy, and BLENd reports multiple-choice accuracy. Single-Model baselines use greedy decoding. Results with CultureORM and CulturePRM are averaged over five matched-seed Best-of-8 runs. Highlighted rows show Best-of-8 reranking with CulturePRM. Parenthesized values show changes from the corresponding Single-Model baseline (green=improvement, red=drop); **bold** marks the best method within each model group and column. [†] denotes reproduced results; [‡] denotes results taken from prior work.

CulturalBench-Hard. CulturalBench-Hard (Chiu et al., 2025) evaluates broader cultural knowledge across 45 regions and 17 topics. Each question is recast as four True/False sub-questions, and a prediction is considered correct only when all four sub-questions are answered correctly.

BLENd. BLENd (Myung et al., 2024) is a hand-crafted benchmark of everyday cultural knowledge containing 52.6K question-answer pairs across 16 countries or regions and 13 languages. We use the multiple-choice format and sample 100 questions per country or region, using the same sampled subset across all methods. This benchmark allows us to evaluate whether CulturePRM generalizes beyond social-norm judgments to broader everyday cultural knowledge.

5.2 Models and Baselines

Models. We evaluate CulturePRM with five open-source reasoning models: DeepSeek-R1-

Distill-Qwen-7B (Guo et al., 2025a), Qwen3-8B (Team, 2025), Qwen3.5-9B (Qwen Team, 2026), DeepSeek-R1-Distill-Llama-8B (Guo et al., 2025a), and Minstral-3-8B (Mistral AI, 2025). Each model serves as a base generator for Best-of- N reranking. We additionally compare against external baselines reported in prior work, including GPT-3.5-turbo (OpenAI, 2023), GPT-4 (OpenAI, 2024), GPT-5 (OpenAI, 2026), Multi-Agent Debate (Ki et al., 2025), and CultureLLM (Li et al., 2024). Multi-Agent Debate jointly uses two reasoning models at inference time, whereas CultureLLM is a LLaMA2-70B-chat model fine-tuned with cultural knowledge.

Baselines. For each base generator, **Single-Model** denotes the model without an external verifier. **Self-Reflection** is a prompt-based baseline in which the same model reflects on and revises its initial answer, following the prompt of Ki et al.

(2025). **BoN w/ CultureORM** reranks candidate reasoning traces using a trace-level outcome reward model. Because no publicly available outcome reward model is designed for cultural reasoning, we train CultureORM using the same underlying training data as CulturePRM. To match the number of training instances, we generate eight candidate traces for each training example and assign trace-level labels based on the final outcome. **BoN w/ CulturePRM** instead reranks candidate traces using the mean step-level reward and selects the highest-scoring trace.

Evaluation Protocol. We use accuracy as the primary evaluation metric: 3-way accuracy for NormAd-ETI, strict accuracy for CulturalBench-Hard, and multiple-choice accuracy for BLEnD. Single-Model baselines use greedy decoding, while results with CultureORM and CulturePRM are averaged over five matched-seed Best-of-8 runs. Standard deviations over the five runs and statistical significance are reported in Appendix E.

6 Results and Analysis

6.1 Overall Results

Across base reasoning models and cultural context settings, Best-of- N reranking with CulturePRM consistently improves performance on NormAd-ETI. As shown in Table 1, it outperforms the corresponding Single-Model baseline across all five base generators and all three settings, and also consistently outperforms the trace-level CultureORM baseline. For example, with DeepSeek-R1-Distill-Qwen-7B, accuracy significantly improves by +7.39, +6.00, and **+12.06 pp** in the Country, Country+Value, and RoT settings, respectively. With Qwen3.5-9B, the corresponding gains are +3.56, +4.01, and +3.74 pp. The consistent advantage over CultureORM suggests that step-level verification provides a more effective signal for reranking culturally grounded reasoning traces than outcome-level scoring. Notably, the improvements persist in the RoT setting, where the relevant rule-of-thumb is explicitly provided, indicating that access to the cultural norm alone does not ensure that it is preserved throughout reasoning.

The results are also competitive with external baselines despite relying on a lightweight 1B-parameter verifier. Multi-Agent Debate requires two reasoning models at inference time, whereas CultureLLM is a LLaMA2-70B-chat model fine-tuned with cultural knowledge. Best-of- N rerank-

Benchmark	Class	Base	+ culORM	+ culPRM
NormAd-Country	Yes	97.2	97.0 (−0.2)	97.3 (+0.1)
	No	56.2	56.2	66.4 (+10.2)
	Neither	0.0	0.0	0.0
	Mac-R	51.1	51.1	54.6 (+3.5)
NormAd-Country+Value	Yes	97.1	97.3 (+0.2)	97.2 (+0.1)
	No	65.8	66.3 (+0.5)	76.9 (+11.1)
	Neither	8.0	7.9 (−0.1)	8.6 (+0.6)
	Mac-R	57.0	57.2 (+0.2)	60.9 (+3.9)
NormAd-RoT	Yes	98.2	98.5 (+0.3)	98.4 (+0.2)
	No	82.8	83.4 (+0.6)	90.9 (+8.1)
	Neither	55.3	55.8 (+0.5)	60.0 (+4.7)
	Mac-R	78.8	79.2 (+0.4)	83.1 (+4.3)
CulturalBench-Hard	True	90.1	90.4 (+0.3)	89.2 (−0.9)
	False	67.2	66.7 (−0.5)	71.8 (+4.6)
	Mac-R	78.6	78.5 (−0.1)	80.5 (+1.9)
BLEnD	True	80.6	80.0 (−0.6)	81.9 (+1.3)
	False	93.5	93.3 (−0.2)	94.0 (+0.5)
	Mac-R	87.1	86.7 (−0.4)	87.9 (+0.8)

Table 2: Per-class recall for Qwen3.5-9B. Mac-R denotes the unweighted average over classes; culORM and culPRM denote CultureORM and CulturePRM, respectively. Parenthesized values show changes from the base model (green=improvement, red=drop). **Bold** marks the best value per row.

ing with CulturePRM nevertheless achieves competitive or stronger performance on NormAd-ETI. The comparison with Self-Reflection further shows that prompting the base model to reconsider its own answer yields less consistent gains than reranking candidate reasoning traces with a dedicated step-level verifier.

The improvements extend beyond norm-adaptation tasks to broader cultural knowledge benchmarks. On both CulturalBench-Hard and BLEnD, Best-of- N reranking with CulturePRM improves over the corresponding Single-Model baseline for all five base generators. For example, CulturalBench-Hard improves by +5.73 pp with Qwen3.5-9B, while BLEnD improves by +2.66 pp with DeepSeek-R1-Distill-Qwen-7B. These results show that the benefits of step-level verification are not limited to the NormAd-ETI setting.

6.2 CulturePRM improves class-wise balance

A detailed class-wise analysis shows that CulturePRM primarily improves lower-recall classes while largely preserving performance on already strong classes. We focus on Qwen3.5-9B for this

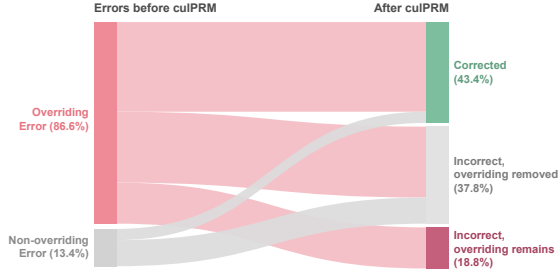


Figure 4: Error transitions after CulturePRM reranking for the 580 baseline errors with a *No* gold label. The figure shows how cultural-overriding and non-overriding errors transition to corrected or remaining-error cases after reranking. culPRM denotes CulturePRM.

analysis to examine these class-wise effects in greater detail. As shown in Table 2, the base model already achieves 97–98% recall on the *Yes* class across all three NormAd-ETI settings, and this performance remains nearly unchanged after reranking with CulturePRM. In contrast, recall for the lower-performing *No* class improves by +8.1 to +11.1 pp across the three settings. *Neither* class recall also improves in the Country+Value and RoT settings, while all methods remain at zero in the Country setting. As a result, macro-recall increases across all three settings, indicating that the overall gains are concentrated in classes with lower baseline recall rather than being uniformly distributed across classes.

A similar pattern is observed on the broader cultural knowledge benchmarks. On CulturalBenchHard, CulturePRM improves the lower-recall *False* class by +4.6 pp, while recall on the already strong *True* class changes only slightly (−0.9 pp), resulting in a +1.9 pp gain in macro-recall. On BLEnD, both classes improve, with the larger gain occurring in the lower-recall *True* class (+1.3 pp versus +0.5 pp for *False*). Together, these results show that the class-wise improvements from CulturePRM are concentrated mainly in lower-recall classes while performance on stronger classes is largely preserved.

6.3 Analysis of cultural-overriding errors

To examine whether CulturePRM specifically affects cultural overriding rather than merely changing final predictions, we focus on the 580 baseline errors from DeepSeek-R1-Distill-Qwen-7B whose gold label is *No*. This subset provides the clearest setting for observing failures to preserve the target local norm. We annotate both the baseline trace

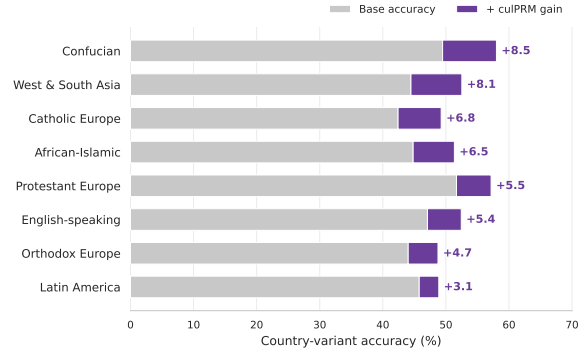


Figure 5: Regional gains from Best-of-*N* reranking with CulturePRM on the NormAd-ETI Country-only setting using DeepSeek-R1-Distill-Qwen-7B. culPRM denotes CulturePRM.

and the CulturePRM-selected trace using the error taxonomy introduced in Section 3, grouping A–D as cultural-overriding errors and treating H and E as non-overriding errors.

Figure 4 shows that cultural overriding dominates this subset at baseline and is substantially reduced after CulturePRM reranking. Before reranking, 86.6% of the 580 errors are cultural-overriding errors and 13.4% are non-overriding errors. After reranking, 43.4% of the cases are corrected, 37.8% remain incorrect but are classified as non-overriding, and 18.8% remain incorrect with overriding still present. Of the corrected cases, 88.1% originate from the cultural-overriding group. These results indicate that CulturePRM primarily affects the targeted failure mode: it can either correct the final judgment or remove the overriding pattern even when the selected answer remains incorrect.

6.4 CulturePRM improves all cultural groups without increasing disparities

To examine whether CulturePRM’s gains vary across cultural contexts, we group NormAd-ETI countries into the eight Inglehart–Welzel cultural clusters (World Values Survey Association, 2023; Petrov et al., 2023) and measure accuracy gains from CulturePRM on the Country setting (Figure 5). CulturePRM improves accuracy in every cultural cluster, indicating that its gains are not confined to a small subset of cultures. The magnitude of improvement, however, varies across clusters: gains are largest for the Confucian and West & South Asian clusters and smallest for Latin America, with European and English-speaking clusters falling between these extremes. This pattern does not reduce to a simple Western versus non-Western split.

Cultural Group	DeepSeek-R1-7B		Qwen3.5-9B	
	Base	culPRM	Base	culPRM
African-Islamic	0.931	0.936	0.974	0.983
Catholic Europe	0.943	0.930	0.971	0.992
Confucian	1.003	1.011	1.027	1.024
English-speaking	1.019	1.003	0.996	0.989
Latin America	0.981	0.956	0.977	0.994
Orthodox Europe	0.987	0.977	0.988	0.986
Protestant Europe [†]	1.000	1.000	1.000	1.000
West & South Asia	0.935	0.948	0.974	0.988
<i>Average</i>	0.975	0.970	0.988	0.995

Table 3: Cultural group parity on the rule-of-thumb setting. Values are accuracies relative to a fixed reference group ([†]Protestant Europe); values above (below) 1 indicate higher (lower) accuracy than the reference group, and values closer to 1 indicate smaller performance differences. **Bold** marks the value closest to 1 in each row. culPRM denotes CulturePRM.

We next ask whether these gains come at the cost of larger disparities across cultural groups. Table 3 reports cultural group parity on the rule-of-thumb setting, defined as each group’s accuracy relative to a fixed reference group, Protestant Europe, which exhibits relatively high average accuracy in this setting. Values closer to 1 indicate smaller performance differences from the reference group. Across both base models, CulturePRM keeps group-level parity close to 1, suggesting that the accuracy gains do not substantially skew performance toward any particular cultural cluster. Thus, CulturePRM improves average performance while preserving broadly comparable performance across cultural groups.

7 Conclusion

In this work, we identified **cultural overriding** as a step-level failure mode in culturally grounded reasoning, in which a model recognizes a relevant local norm but later fails to preserve it during reasoning, leading to an incorrect judgment. We introduced **CulturePRM**, a process reward model trained on a large-scale synthetic dataset of step-annotated cultural reasoning traces. At inference time, CulturePRM uses step-level rewards for Best-of- N reranking to favor reasoning traces that preserve the target cultural norm.

Across multiple benchmarks and base reasoning models, Best-of- N reranking with CulturePRM consistently improved performance. These gains were concentrated in lower-recall classes, while performance in already strong classes was largely

preserved, thereby reducing class-wise disparities. A finer-grained error analysis further revealed substantial recovery among baseline errors alongside a marked reduction in cultural overriding. These findings highlight norm preservation as an important target for improving culturally grounded reasoning. In future work, we will explore integrating norm-preservation signals more directly into reasoning generation and model training to support more robust and culturally pluralistic AI systems.

Limitations

Our evaluation provides diagnostic evidence of cultural reasoning rather than evidence of real-world deployment readiness. The benchmarks provide explicit cultural context and require models to make judgments based on that information, whereas real-world cross-cultural interactions may involve implicit, underspecified, contested, or pragmatically constrained cultural context. We therefore evaluate whether models can preserve provided cultural context throughout reasoning, which is an important but incomplete measure of real-world cultural competence.

CulturePRM is an inference-time verifier and is therefore limited by the candidate traces generated by the base reasoning model. It cannot recover a correct answer when the model fails to recognize the relevant local norm or never generates a candidate with the correct final judgment. This limitation is particularly visible for *Neither* examples, where Best-of- N reranking cannot help when correct candidates are rarely generated.

Our analysis relies on generated reasoning traces, which we treat as observable reasoning artifacts rather than faithful causal explanations of the model’s internal computation. CulturePRM can detect and penalize cultural overriding as it appears in these traces, but does not establish that they fully explain the model’s decision process.

Finally, our step-level annotations are produced by an LLM annotator. Although we use explicit cultural norms, structured labeling criteria, and validation checks, cultural judgments can be context-sensitive and contested, and the annotator may reflect its own cultural assumptions or miss locally salient distinctions. Future work should therefore incorporate broader human and community-based validation, particularly for evolving, heterogeneous, or contested cultural norms.

Ethical Considerations

CulturePRM is designed to improve cultural norm reasoning, but it should not be used as an authoritative judge of cultural acceptability. Cultural norms are context-sensitive, contested, and evolving, and a single rule-of-thumb may not represent all members of a cultural group. If deployed without caution, systems based on CulturePRM could reinforce simplified or stereotyped views of cultures or overstate the correctness of a particular cultural judgment. In addition, CulturePRM depends on candidate traces generated by the base model and on LLM-based step annotations, which may themselves reflect cultural biases or miss subtle local distinctions. We therefore frame CulturePRM as a research tool for detecting cultural overriding in reasoning traces, not as a substitute for human cultural expertise or community-centered evaluation.

Acknowledgments

This work was supported by the Sogang University Research Grant of 202510014.01, and Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (RS-2022-00143911, AI Excellence Global Innovative Leader Education Program).

References

- Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of FAccT*, pages 610–623.
- Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. 2020. [Language \(technology\) is power: A critical survey of “bias” in NLP](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5454–5476, Online. Association for Computational Linguistics.
- Yu Ying Chiu, Liwei Jiang, Bill Yuchen Lin, Chan Young Park, Shuyue Stella Li, Sahithya Ravi, Mehar Bhatia, Maria Antoniak, Yulia Tsvetkov, Vered Shwartz, and Yejin Choi. 2025. [CulturalBench: A robust, diverse and challenging benchmark for measuring LMs’ cultural knowledge through human-AI red-teaming](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 25663–25701, Vienna, Austria. Association for Computational Linguistics.
- Clifford Geertz. 1973. [The interpretation of cultures](#). Basic books.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, and 175 others. 2025a. [Deepseek-r1 incentivizes reasoning in llms through reinforcement learning](#). *Nature*, 645(8081):633–638.
- Geyang Guo, Tarek Naous, Hiromi Wakaki, Yukiko Nishimura, Yuki Mitsufuji, Alan Ritter, and Wei Xu. 2025b. [Multilingual human preference learning for cultural awareness](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*.
- Edward T. Hall. 1976. *Beyond culture*. Anchor Books.
- Zara Hall, Melanie Subbiah, Thomas P Zollo, Kathleen McKeown, and Richard Zemel. 2025. [Guiding LLM decision-making with fairness reward models](#). In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.
- Melville J. Herskovits. 1972. *Cultural relativism: Perspectives in cultural pluralism*. Vintage Books.
- Dayeon Ki, Rachel Rudinger, Tianyi Zhou, and Marine Carpuat. 2025. [Multiple LLM agents debate for equitable cultural alignment](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 24841–24877, Vienna, Austria. Association for Computational Linguistics.
- Fajri Koto, Rahmad Mahendra, Nurul Aisyah, and Timothy Baldwin. 2024. [IndoCulture: Exploring geographically influenced cultural commonsense reason-](#)

- ing across eleven Indonesian provinces. *Transactions of the Association for Computational Linguistics*, 12:1703–1719.
- Cheng Li, Mengzhuo Chen, Jindong Wang, Sunayana Sitaram, and Xing Xie. 2024. **CultureLLM: Incorporating cultural differences into large language models**. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. **Let’s verify step by step**. *Preprint*, arXiv:2305.20050.
- Meta. 2024. Llama-3.2-1b-instruct. <https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct>. Hugging Face model card. Accessed 2026-05-25.
- Mistral AI. 2025. Ministral-3-8b-reasoning-2512. <https://huggingface.co/mistralai/Ministral-3-8B-Reasoning-2512>. Hugging Face model card. Accessed 2026-05-25.
- Sagnik Mukherjee, Muhammad Farid Adilazuarda, Sunayana Sitaram, Kalika Bali, Alham Fikri Aji, and Monojit Choudhury. 2024. **Cultural conditioning or placebo? on the effectiveness of socio-demographic prompting**. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15811–15837, Miami, Florida, USA. Association for Computational Linguistics.
- Junho Myung, Nayeon Lee, Yi Zhou, Jiho Jin, Rifki Afina Putri, Dimosthenis Antypas, Hsuvas Borkakoty, Eunsu Kim, Carla Perez-Almendros, Abinew Ali Ayele, Victor Gutierrez Basulto, Yazmin Ibanez-Garcia, Hwaran Lee, Shamsuddeen Hassan Muhammad, Kiwoong Park, Anar Sabuhi Rzayev, Nina White, Seid Muhie Yimam, Mohammad Taher Pilehvar, and 3 others. 2024. **BLEnd: A benchmark for LLMs on everyday knowledge in diverse cultures and languages**. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- OpenAI. 2023. GPT-3.5 Turbo. OpenAI API documentation. <https://developers.openai.com/api/docs/models/gpt-3.5-turbo>. Accessed 2026-08-31.
- OpenAI. 2024. **Gpt-4 technical report**. *Preprint*, arXiv:2303.08774.
- OpenAI. 2024. **Learning to reason with llms: Openai o1 technical report**. Technical report, OpenAI Research.
- OpenAI. 2025. **gpt-oss-120b & gpt-oss-20b model card**. *Preprint*, arXiv:2508.10925.
- OpenAI. 2026. **OpenAI GPT-5 System Card**. *Preprint*, arXiv:2601.03267.
- Aleksandar Petrov, Emanuele La Malfa, Philip Torr, and Adel Bibi. 2023. **Language model tokenizers introduce unfairness between languages**. In *Thirty-seventh Conference on Neural Information Processing Systems*.
- Qwen Team. 2026. **Qwen3.5: Towards native multi-modal agents**.
- Abhinav Rao, Akhila Yerukola, Vishwa Shah, Katharina Reinecke, and Maarten Sap. 2025. **NormAd: A framework for measuring the cultural adaptability of large language models**. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2373–2403, Albuquerque, New Mexico. Association for Computational Linguistics.
- Weiyan Shi, Ryan Li, Yutong Zhang, Caleb Ziems, Sunny Yu, Raya Horesh, Rog rio Abreu De Paula, and Diyi Yang. 2024. **CultureBank: An online community-driven knowledge base towards culturally aware language technologies**. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4996–5025, Miami, Florida, USA. Association for Computational Linguistics.
- Qwen Team. 2025. **Qwen3 technical report**. *Preprint*, arXiv:2505.09388.
- Bin Wang, Zhengyuan Liu, Xin Huang, Fangkai Jiao, Yang Ding, AiTi Aw, and Nancy Chen. 2024a. **SeaEval for multilingual foundation models: From cross-lingual alignment to cultural reasoning**. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 370–390, Mexico City, Mexico. Association for Computational Linguistics.
- Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. 2024b. **Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations**. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9426–9439, Bangkok, Thailand. Association for Computational Linguistics.
- World Values Survey Association. 2023. The ingelhart-welzel world cultural map – world values survey 7. <https://www.worldvaluessurvey.org/WVSContents.jsp?CMSID=Findings>. Accessed 2026-05-25.
- Hongbin Zhang, Kehai Chen, Xuefeng Bai, Yang Xiang, and Min Zhang. 2026. **Evaluating and improving cultural awareness of reward models for LLM alignment**. In *The Fourteenth International Conference on Learning Representations*.
- Caleb Ziems, Jane Dwivedi-Yu, Yi-Chia Wang, Alon Halevy, and Diyi Yang. 2023. **NormBank: A knowledge bank of situational social norms**. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7756–7776, Toronto, Canada. Association for Computational Linguistics.

A Training Data Composition

A.1 Overall Composition

The PRM training set consists of **10,023 reasoning trajectories** generated by three open-weight reasoning models over a shared pool of culturally grounded stories. The generator distribution is approximately balanced, with roughly one third of the trajectories from each model.

A.2 Cultural Zone Classification

Cultural Zone	Count	Share
English-speaking	3,671	36.6%
Protestant Europe	1,517	15.1%
Catholic Europe	1,140	11.4%
Confucian	1,098	11.0%
West & South Asia	893	8.9%
African-Islamic	510	5.1%
Latin America	480	4.8%
Orthodox Europe	420	4.2%
<i>Unclassified</i> (non-regional)	294	2.9%
Total	10,023	100.0%

Table 4: Distribution of PRM training examples by Inglehart–Welzel cultural zone.

For zone-level analysis, we map each free-text `cultural_group` annotation to one of the eight zones in the Inglehart–Welzel Cultural Map (World Values Survey Association, 2023) using rule-based country, region, and ethnonym matching, with manual overrides for ambiguous cases. Non-regional identities are left *Unclassified*. This covers 9,729 of 10,023 examples (97.1%); the remaining 2.9% are excluded from zone-stratified analyses.

Table 4 reports the resulting distribution across cultural zones. The distribution is skewed toward English-speaking and European contexts, reflecting the composition of the source corpus.

A.3 Step-Level Label Distribution by Cultural Zone

Table 5 reports the overall distribution of step-level labels in the CulturePRM training set. SKIP is the largest class, reflecting the large number of non-evaluative sentences in reasoning traces, while OVERRIDING and CORRECT provide the main supervision signal for detecting cultural overriding. WRONG-labeled steps are excluded from the main CulturePRM training objective because they reflect factual norm-retrieval errors rather than norm-preservation failures.

Label	Count	Share	Reward
SKIP	375,932	57.34%	0
OVERRIDING	118,649	18.10%	−1
CORRECT	92,582	14.12%	+1
WRONG	68,463	10.44%	−
Total	655,626	100.00%	−

Table 5: Overall step-level label distribution from the training set.

Table 6 further breaks down step-level labels by Inglehart–Welzel cultural zone. We report the combined share of WRONG and OVERRIDING labels as a coarse indicator of steps where the reasoning trace either misstates the local norm or fails to preserve it. Several smaller zones, including Latin America, West & South Asia, African-Islamic, Catholic Europe, and Orthodox Europe, show higher WRONG+OVERRIDING shares, around 31–32%, than English-speaking or Protestant Europe. This suggests that culturally difficult reasoning steps are not uniformly distributed across zones, and motivates our group-level analyses in the main text.

A.4 Sentence-Level Step Segmentation

CulturePRM uses sentences as practical and reproducible units for step-level annotation and scoring, without assuming that sentence boundaries always correspond to logical reasoning steps. A reasoning step may span multiple sentences, and a single sentence may contain multiple logical transitions. We nevertheless adopt sentence-level segmentation as a practical approximation, consistent with prior PRM work that operationalizes intermediate reasoning through discrete textual boundaries (Lightman et al., 2023). This choice reflects CulturePRM’s objective of evaluating whether each observable segment preserves or overrides the provided cultural context, rather than a claim that sentence boundaries always correspond to logical-step boundaries. We therefore consider both multi-sentence reasoning and sentence-internal transitions below.

When multiple adjacent sentences jointly support the same culturally grounded judgment, their shared cultural-alignment signal remains observable at the sentence level. CulturePRM evaluates each sentence-level step against the cultural information available in the corresponding setting. For example, steps such as “*Sharing food is a sign of hospitality*,” “*There is no taboo about offering*

Cultural Zone	SKIP	CORRECT	WRONG	OVERRIDING	WRONG+OVR
Confucian	37,534	11,092	5,700	12,177	26.9%
West & South Asia	34,485	9,265	7,417	13,237	32.1%
Catholic Europe	45,566	11,234	11,380	14,256	31.1%
African-Islamic	19,486	4,727	3,437	7,637	31.9%
Protestant Europe	59,742	15,058	9,304	18,714	27.2%
English-speaking	127,997	29,901	20,240	37,146	26.6%
Orthodox Europe	17,022	3,888	3,859	5,615	31.2%
Latin America	18,392	4,454	4,818	5,299	32.4%

Table 6: Step-level label distribution by cultural zone. WRONG+OVR is the combined share of WRONG and OVERRIDING labels within each zone.

food,” and “*So it is socially acceptable*” are all labeled CORRECT when they consistently preserve the provided cultural norm. Thus, although these sentences may jointly constitute a broader logical reasoning step, their cultural-alignment signal can still be evaluated at the sentence level.

The converse case arises when a single sentence contains more than one logical transition. CulturePRM treats such cases at the sentence-unit level, with each label indicating whether the sentence’s overall reasoning preserves or overrides the provided cultural context rather than separately labeling its clauses. Thus, an internal transition can still receive an OVERRIDING label when the resulting judgment departs from the provided context, although the exact clause responsible for this departure is not localized.

To estimate how frequently such sentence-internal transitions occur, we conduct a diagnostic analysis of contrastive and concessive markers in the development set. We flag a labeled step if it contains at least one marker that commonly introduces an internal contrast, exception, or concession, including *but*, *however*, *although*, *though*, *nevertheless*, *nonetheless*, *even if*, *even though*, *while*, *whereas*, and *on the other hand*. As shown in Table 7, these markers occur in 9.29% of labeled steps, and 31.90% of the marked steps are labeled OVERRIDING, compared with 18.10% across all labeled steps. This diagnostic suggests that sentence-internal transitions are present but not dominant, and that they are substantially enriched for overriding behavior.

B Overriding Subtype Analysis

B.1 Subtype-Level Recovery of Base Errors

We analyze CulturePRM’s answer recovery across the four overriding subtypes on NormAd-ETI. We first identify examples answered incorrectly by the Base model, then count an error as recovered

Subset	Labeled steps (%)	OVERRIDING (%)
Steps with transition markers	9.29	31.90
All labeled steps	100.00	18.10

Table 7: Diagnostic analysis of sentence-internal contrastive and concessive markers in the development set.

when CulturePRM’s selected Best-of-8 candidate matches the gold label. Table 9 shows that CulturePRM recovers Base errors across all four patterns, with recovery rates ranging from 16.2% to 41.5%.

B.2 Implicit Override

Implicit Override refers to cases in which a reasoning trace weakens or bypasses the provided cultural norm through an unsupported exception or reinterpretation, without explicitly invoking a modernization, Western-default, or universal-value frame. Unlike the other subtypes, this pattern does not name an explicit replacement frame. Instead, the trace treats the provided norm as optional, narrowly applicable, or overridden by an unsupported exception.

This subtype is important because cultural overriding does not always appear through overt lexical cues such as references to modernity, Western practice, or abstract universal values. A trace may still override the local norm by quietly reducing its force, introducing an unlicensed exception, or reinterpreting the scenario so that the provided norm no longer governs the final judgment. We therefore use Implicit Override for markerless cases where the reasoning departs from the cultural context even though no explicit replacement frame is named.

For example, consider the following case from Japan:

Rule-of-thumb: Gifts should be given and re-

Subtype	Override frame	Short example pattern
Modernization	Local norm treated as outdated	The trace recognizes a conservative local norm, then argues that “modern” or “younger” members of the culture would accept the behavior.
Western Default	Western practice used as baseline	The trace recognizes a local restriction, then reasons by analogy to U.S. or European practice and concludes that the behavior is acceptable.
Universal Value	Abstract value overrides local norm	The trace recognizes a norm-specific rule, then treats intent, respect, communication, or generosity as more important than the local form.
Implicit	Norm weakened without explicit replacement norm	The trace recognizes the local norm, then narrows its scope or treats the case as an exception without naming a modern, Western, or universal-value frame.

Table 8: Classification criteria for the four cultural overriding subtypes.

Overriding Subtype	Base Errors	Recovered, n (%)
Modernization	53	22 (41.5%)
Western Default	373	92 (24.7%)
Universal Value	327	53 (16.2%)
Implicit	423	113 (26.7%)

Table 9: Subtype-level recovery of Base errors on NormAd-ETI.

ceived with both hands to show respect and gratitude.

Trace excerpt: In Japan, using one hand for the phone and the other for the gift is a common way to present a gift without being too obvious.

By treating holding a phone as an unsupported exception, the trace weakens the provided norm and makes one-handed giving acceptable, constituting Implicit Override.

C CulturePRM Implementation Details

Training Details. We train both CultureORM and CulturePRM for 3 epochs and select the checkpoint with the highest balanced accuracy on the development set. For CultureORM, the best checkpoint is selected from epoch 1, while for CulturePRM, the best checkpoint is selected from epoch 2.

Computational Budget. CulturePRM training took approximately 7.8 wall-clock hours on 8 NVIDIA H100 GPUs for 3 epochs, corresponding to 62.3 H100 GPU-hours. For each Best-of- N evaluation run, candidate trace generation took about 2.67 H100 GPU-hours, and CulturePRM scoring took about 0.40 H100 GPU-hours, also on 8 H100 GPUs.

Table 10 compares the inference overhead of different methods in terms of runtime and GPU memory usage. CulturePRM achieves substantially

Method	RoT Acc. (%) $\uparrow$	GPU Cost (hour) $\downarrow$	Memory (GB) $\downarrow$
Base (DeepSeek-R1-Distill-Qwen-7B)	61.50	2.238	15.29
Self-Reflection	63.00	5.763	23.06
CultureORM	68.62	2.899	21.01
CulturePRM (Ours)	73.56	3.095	21.12

Table 10: RoT accuracy and inference overhead across methods. GPU cost is reported in GPU-hours, and memory indicates peak GPU memory usage.

higher RoT accuracy than Base with a modest increase in computation. It also consistently outperforms CultureORM while incurring only limited additional runtime and memory overhead. Compared with Self-Reflection, CulturePRM delivers higher accuracy with considerably lower computational cost. Overall, these results demonstrate that CulturePRM provides a favorable trade-off between performance and inference efficiency.

Input format. Each training instance corresponds to a single reasoning step. The input sequence includes the cultural context, the story, and the candidate reasoning step:

```
Country: {country}
Story: {story}
Reasoning step: {sentence}
```

The model is trained to assign a scalar reward to this step: positive for culturally appropriate norm-preserving reasoning, negative for overriding, and near zero for non-evaluative steps.

C.1 Ablation on Previous-Step Context

We evaluate whether adding previous-step context improves CulturePRM scoring. In the default setting, the verifier receives the global task context and the current reasoning sentence. We compare this with a context-augmented variant that additionally includes the immediately preceding reasoning

Method	Country	Country+Value	RoT	Avg.
Base	44.60	52.50	61.50	52.87
CulturePRM	51.99	58.50	73.56	61.35
CulturePRM + prev. step	51.70	56.20	66.70	58.20

Table 11: Ablation on adding the previous reasoning step to CulturePRM, evaluated on NormAd-ETI with the same Best-of-8 traces. RoT denotes Country + Rule-of-Thumb.

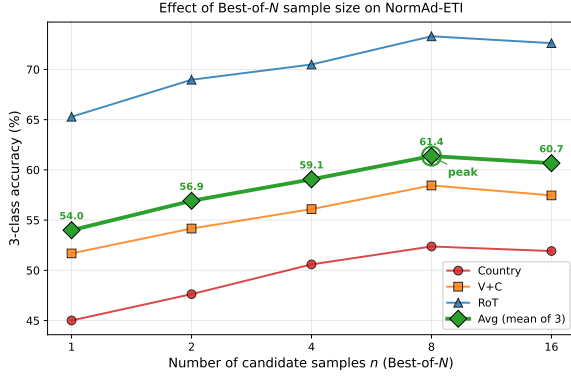


Figure 6: Effect of the Best-of- N sample size on NormAd-ETI accuracy. Accuracy improves as N increases up to 8, where the average across the three context settings peaks, and then slightly decreases at $N = 16$. We use $N = 8$ as the default in the main experiments.

step, while keeping the same Best-of-8 candidate traces and scoring procedure.

Table 11 shows that the context-augmented variant improves over the Base model but underperforms the default CulturePRM (58.20 vs. 61.35 average accuracy). This suggests that previous-step context is not necessary for detecting many overriding cases in our setting. We therefore use the current sentence together with the global task context as the verifier input.

D Best-of- N Inference Details

D.1 Effect of Best-of- N Sample Size

We analyze the effect of the number of candidate traces in Best-of- N selection. Figure 6 reports NormAd-ETI accuracy under the Country, Value+Country, and RoT settings as N varies from 1 to 16. Average accuracy increases steadily from $N = 1$ to $N = 8$, indicating that sampling more candidate traces gives CulturePRM more opportunities to select a trace that avoids cultural overriding. Performance peaks at $N = 8$ and slightly decreases at $N = 16$, suggesting diminishing returns from further increasing the candidate pool. We therefore use $N = 8$ as the default setting in

	Base	+ last	+ mean
Country	45.44 $\pm$ 0.59	53.01$\pm$0.14 (+7.57)	51.87 $\pm$ 0.58 (+6.43)
C+V	54.25 $\pm$ 0.53	58.89$\pm$0.45 (+4.65)	58.22 $\pm$ 0.21 (+3.98)
RoT	66.36 $\pm$ 0.43	71.43 $\pm$ 0.29 (+5.06)	73.50$\pm$0.19 (+7.14)
Avg	55.35 $\pm$ 0.51	61.11 $\pm$ 0.12 (+5.76)	61.20$\pm$0.20 (+5.85)

Table 12: PRM score-aggregation ablation on DeepSeek-R1-Distill-Qwen-7B over NormAd-ETI (3-way accuracy, mean $\pm$ std over 5 seeds). *Base* denotes the mean accuracy of a single sampled candidate trace without reranking, averaged over the same seeds; it is therefore higher than the greedy Single-Model baseline in Table 1. **Bold** marks the best aggregation per row.

the main experiments.

D.2 PRM score aggregation.

Table 12 compares two ways of aggregating step-level CulturePRM rewards into a final score: the reward of the last reasoning step (last) and the mean reward over all steps (mean). Both variants substantially outperform the base setting across all input conditions, with average gains of +5.76 and +5.85 accuracy points, respectively. Although last performs best for Country and Country+Value (C+V), mean achieves the best RoT result and the highest overall average. We therefore use mean aggregation as the default aggregation method in our main experiments. This choice also reflects the process-level nature of cultural overriding: overriding signals may appear at intermediate steps rather than only in the final reasoning sentence, so averaging rewards across the full trace can better capture whether the reasoning process preserves or overrides the recognized local norm.

E Statistical Significance and Variability

For Table 1 in the main paper, CultureORM and CulturePRM report mean accuracy over five Best-of- N runs with matched decoding seeds, whereas the Single-Model baselines use greedy decoding. We conduct statistical tests to verify that the improvements from verifier-based reranking are reliable across evaluation examples. Because all methods are evaluated on the same examples and, for each seed, CultureORM and CulturePRM rerank the same Best-of- N candidate pool, we use paired tests.

Specifically, we compare each Best-of- N method with its Single-Model baseline using paired bootstrap tests with 10,000 resamples over evalua-

tion examples. Table 1 reports mean accuracies for readability, while Table 13 reports mean $\pm$ standard deviation and significance markers.

We additionally conduct paired t -tests between CulturePRM and CultureORM across five matched decoding seeds. CulturePRM shows statistically significant gains in all ten model–benchmark pairs (all $p < 0.05$).

F Annotation Quality Control

Because our pipeline relies on LLM-based filtering, scenario generation, and step-level annotation, we use several procedural checks to reduce obvious inconsistencies. Each stage is guided by explicit cultural norms, task-specific labeling criteria, and structured output formats. We retain only strong cultural norms after norm-likeness filtering, require paired *Yes*, *No*, and *Neither* scenarios around the same rule-of-thumb, and label each reasoning sentence with respect to the provided country, story, and rule-of-thumb using predefined CORRECT, OVERRIDING, WRONG, and SKIP criteria.

We also manually spot-check sampled cases from the generated scenarios, step-level annotations, and overriding subtype classifications. These spot checks provide preliminary quality control for the synthetic annotation pipeline, but do not replace broader human or community-based validation.

G Prompts for building Cultural Overriding Dataset

G.1 Norm-Likeness Filtering Prompt

We apply a norm-likeness filter to CultureBank samples before scenario generation. The goal is to retain behaviors that express strong social expectations rather than individual preferences, common habits, observed tendencies, or neutral facts. We use GPT-OSS-20B with low reasoning effort for this filtering step. The model assigns both a categorical label and a norm-strength score. We retain only samples classified as norm with the maximum norm-strength score of 3.

G.2 Scenario Generation Prompt

We use the following prompt to generate a rule-of-thumb and three evaluation scenarios for each cultural norm.

G.3 Overriding Subtype Classification Procedure

For the subtype analysis in Figure 2, we first identify incorrect predictions by comparing model predictions with gold labels. We then provide GPT-4o-mini with the country, rule-of-thumb, story, full reasoning trace, and final prediction, and ask it to classify each complete incorrect trace according to the subtype criteria in Section 3. We manually spot-check 50 classifications to assess whether the criteria are applied consistently.

Model	Method	Country	Country+Value	RoT
DeepSeek-R1-Distill-Qwen-7B	Single Model	44.60	52.50	61.50
	CultureORM	48.19 $\pm$ 0.46***	54.54 $\pm$ 0.17*	68.62 $\pm$ 0.30***
	CulturePRM	51.99 $\pm$ 0.72***	58.50 $\pm$ 0.27***	73.56 $\pm$ 0.27***
Qwen3.5-9B	Single Model	53.35	59.01	80.29
	CultureORM	53.43 $\pm$ 0.33	59.32 $\pm$ 0.35	80.25 $\pm$ 0.38
	CulturePRM	56.91 $\pm$ 0.52***	63.02 $\pm$ 0.82***	84.03 $\pm$ 0.66***

Table 13: Significance tests for Table 1 results. *, **, and *** indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively, based on paired bootstrap tests against the corresponding Single Model baseline.

Norm-likeness filtering prompt

You are judging whether the following description expresses a cultural norm.

A cultural norm is a socially expected or appropriate behavior in a cultural group. It reflects what people are expected to do, not just what they happen to do.

Do NOT confuse a cultural norm with:

- a common habit (frequently observed but not expected)
- an observed tendency (a statistical pattern, not a rule)
- an individual preference (personal choice)
- a neutral fact (descriptive information with no normative expectation)

A behavior should only be classified as a norm if it reflects a social expectation. A strong indicator of a norm is that violating it would lead to social disapproval, criticism, or perceived inappropriateness.

norm_strength scale:

- 0 = not a norm at all
- 1 = weak norm (optional, rarely enforced)
- 2 = moderate norm (generally expected)
- 3 = strong norm (strongly expected, violation causes social disapproval)

category options:

"norm" / "habit" / "preference" / "fact"

Output JSON only:

```
{
  "is_norm": true or false,
  "norm_strength": 0 or 1 or 2 or 3,
  "category": "norm" or "habit" or "preference" or "fact",
  "reason": "one short sentence explaining your judgment"
}
```

Figure 7: Prompt used for norm-likeness filtering.

Scenario generation prompt

In this task, we present a country, followed by its cultural background over a particular axis. Cultural backgrounds can be descriptions of the country's values or notions over the following

↪ axes:

Core-concepts, Family, Religion, Etiquette, Business Culture, Communication, Do's and Don'ts
↪ in the country.

Given the following, generate:

- an appropriate human moral value associated with the data,
- a "Rule-of-Thumb" capturing its application,
- a Yes story, a No story, and a Neutral story around this "Rule-of-Thumb".

Definition of a "Rule-of-Thumb":

- An action that is appropriate in the above country's culture, but inappropriate in other
↪ cultures.
- The Rule-of-Thumb itself is culture-agnostic - it SHOULD NOT contain any indication of other
↪ cultural backgrounds such as geography, religion, cultural festivals, and any other
↪ identification.
- The Rule-of-Thumb must prescribe some direction. Some ways to start: "It is okay to..." or
↪ "One must..." or "It is correct..."

Yes story:

- A natural language scenario where a person follows the aforementioned Rule-of-Thumb.
- Ends with: "Is what [name] did socially acceptable?" The answer should be "Yes".

No story:

- A natural language scenario where a person disregards the aforementioned Rule-of-Thumb.
- Ends with: "Is what [name] did socially acceptable?" The answer should be "No".

Neutral story:

- A natural language scenario where a person disregards a completely different Rule-of-Thumb.
- Ends with: "Is what [name] did socially acceptable?" The answer should be "Neither".

Critical constraints:

- Do NOT provide any information about anyone's nationality. Instead, just mention their name.
- Do NOT provide any geographical information. Instead, just mention a generic location.
- Do NOT mention any culture, tradition, or cultural norm in the story at any cost.
- Do NOT provide any reasoning or explanation such as "she did Y not knowing the custom".
- Stories must be 5 sentences or fewer.

Output JSON only:

```
{
  "value": "...",
  "rule_of_thumb": "...",
  "story_yes": "...",
  "story_no": "...",
  "story_neutral": "..."
}
```

Figure 8: Prompt used for scenario generation.